



CS670: Cryptographic Techniques for Privacy Preservation

Module 4: Private Information Retrieval

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A few talking points

Assignments

Quiz 3 and 4 on November 3 and November 7

CS714 being offered next semester

UGPs and M.Tech/PhD Thesis

Course Feedback

Private Information Retrieval



Downloader



Wants to watch a movie on Netflix.

However, she does not want Netflix to know what she watched.

Private Information Retrieval

Can we download a record from a remote database server, without the server learning what we downloaded?

Yes! (under certain assumptions, though)

Private Information Retrieval

Private Information Retrieval is a cryptographic technique that allows a user to download from a remote database without the database owner learning what was downloaded!

In general, the user can learn more than what they asked for.

If we have an additional condition that the user is supposed to learn only the record they requested, we refer to it as Symmetric Private Information Retrieval (SPIR).

Private Information Retrieval

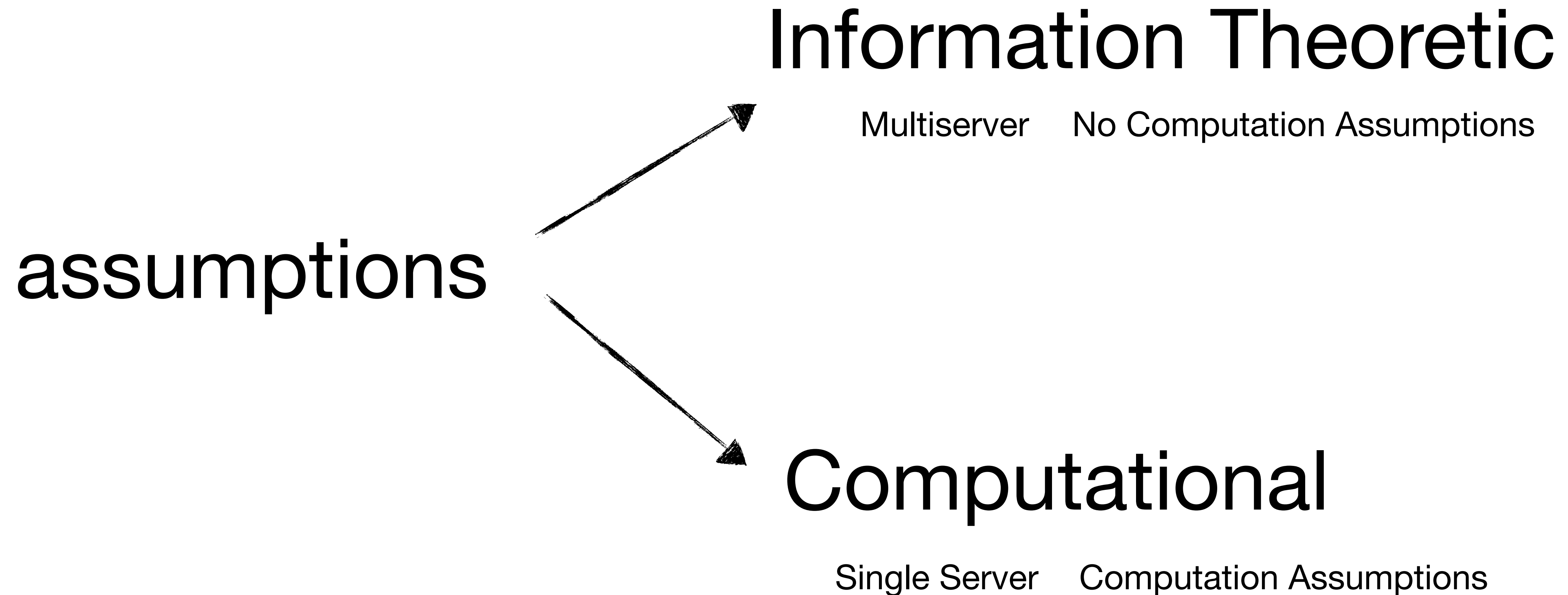
What is the difference between Private Information Retrieval and Anonymous Information Retrieval?

Tor is an example of anonymous information retrieval!

Some other examples where PIR is useful?

You may want to look up something in a patent database!

Private Information Retrieval



Simplest PIR Protocol

What is the simplest PIR protocol?

Naively downloading the entire database!



Downloader

I want to watch a video

Here is the entire Netflix Catalogue



Is this SPIR?

Private Information Retrieval

Information Theoretic Private Information Retrieval (IT-PIR)

Multiple PIR Servers

Each holding the replica of database



(a matrix of 0s and 1s)

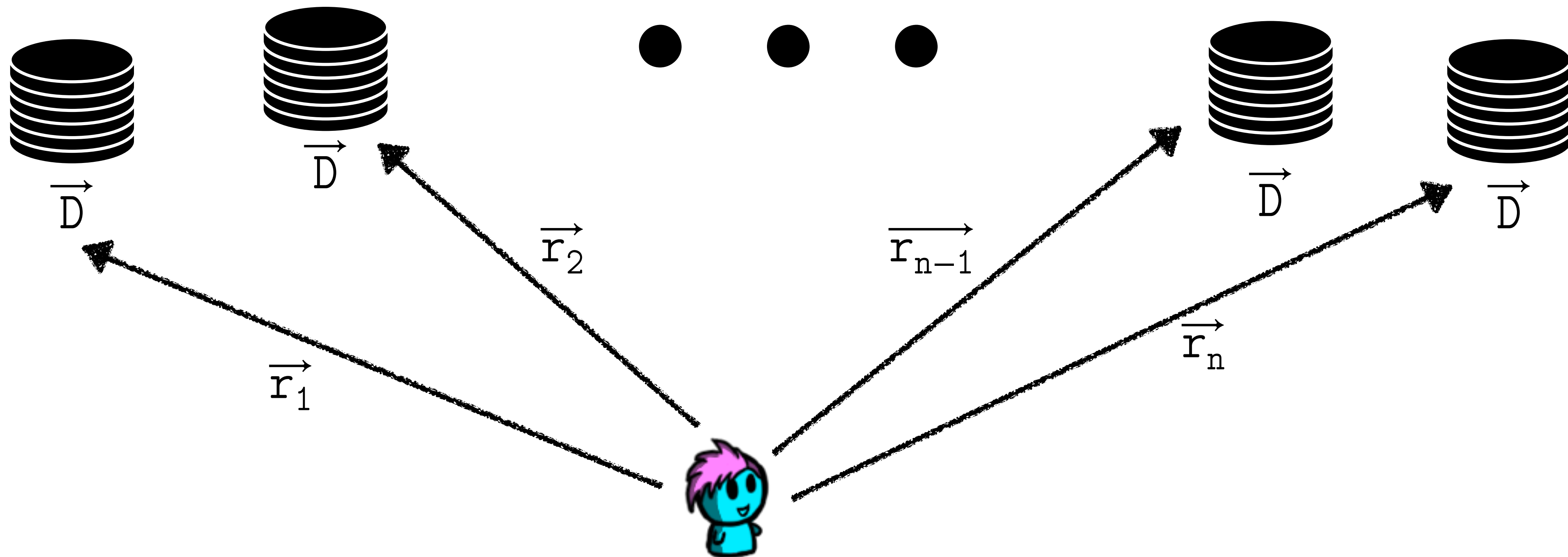
Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

IT-Private Information Retrieval

(a matrix of 0s and 1s)

Multiple PIR Servers Each holding the replica of database

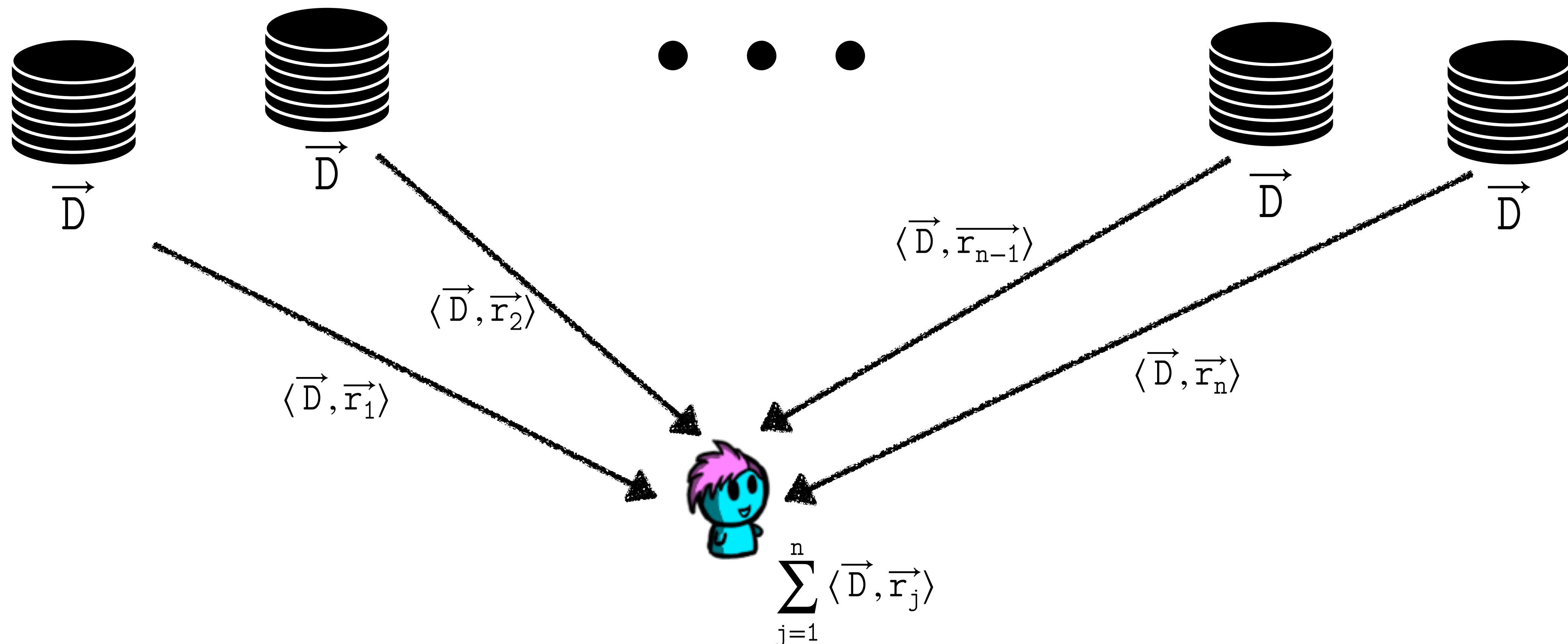
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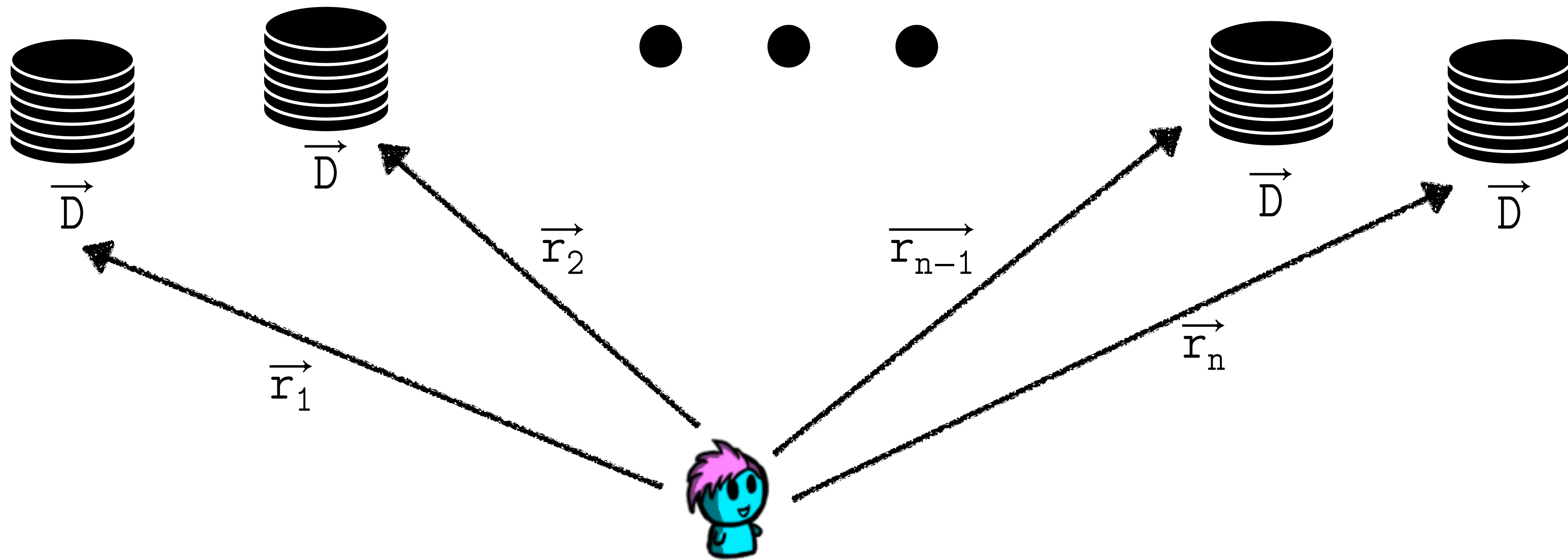
IT-Private Information Retrieval *(a matrix of 0s and 1s)*

Multiple PIR Servers Each holding the replica of database

Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.



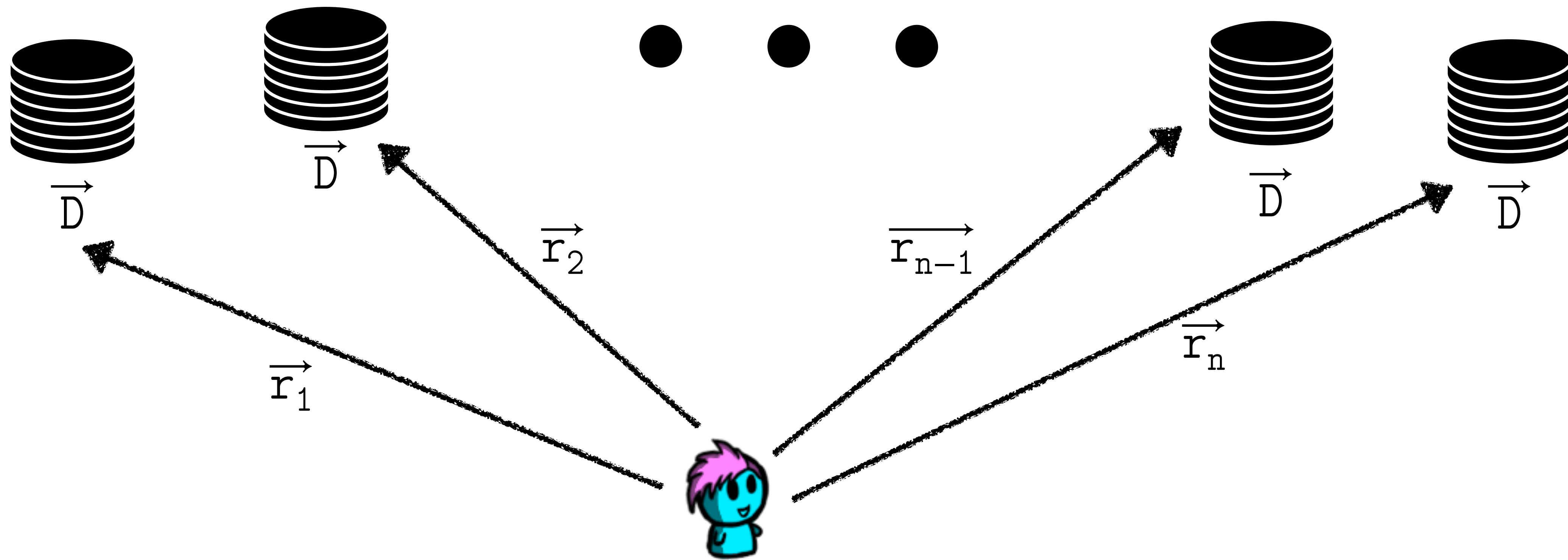
IT-Private Information Retrieval



Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

What should be the property of these vectors? $\sum_{j=1}^n \vec{r}_j = \vec{e}_i$

IT-Private Information Retrieval

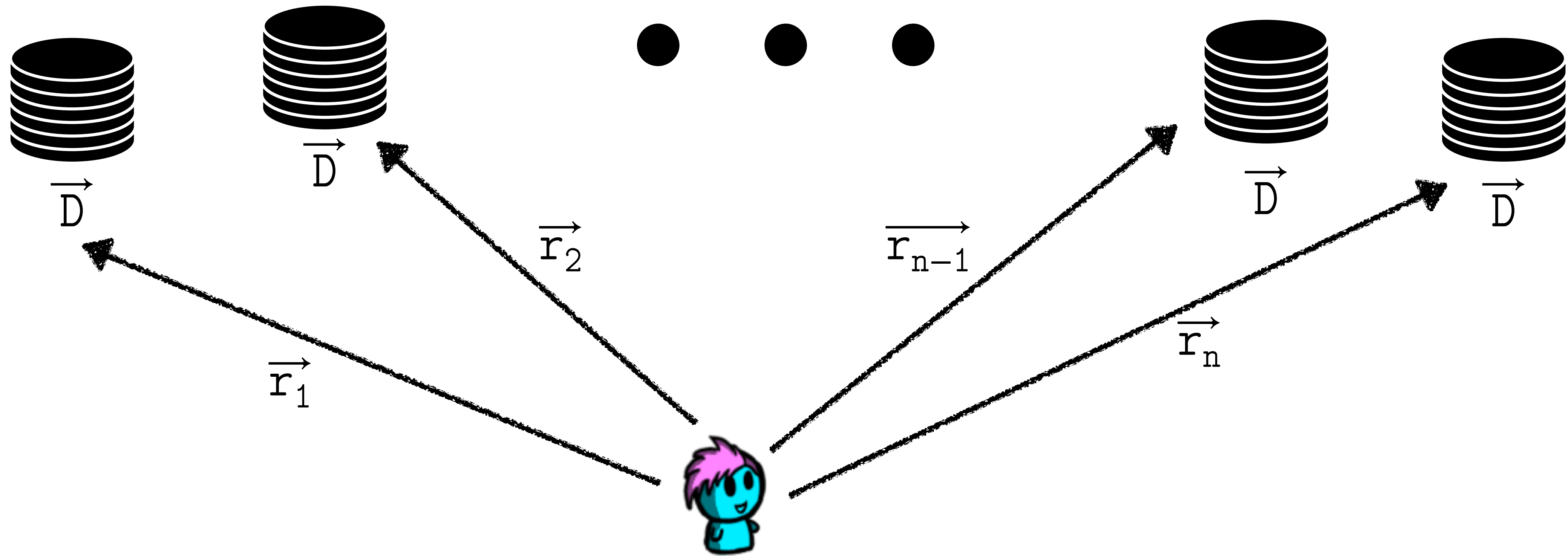


Goal: To retrieve the j^{th} row (record) of the database (matrix) privately.

What is the upload cost?

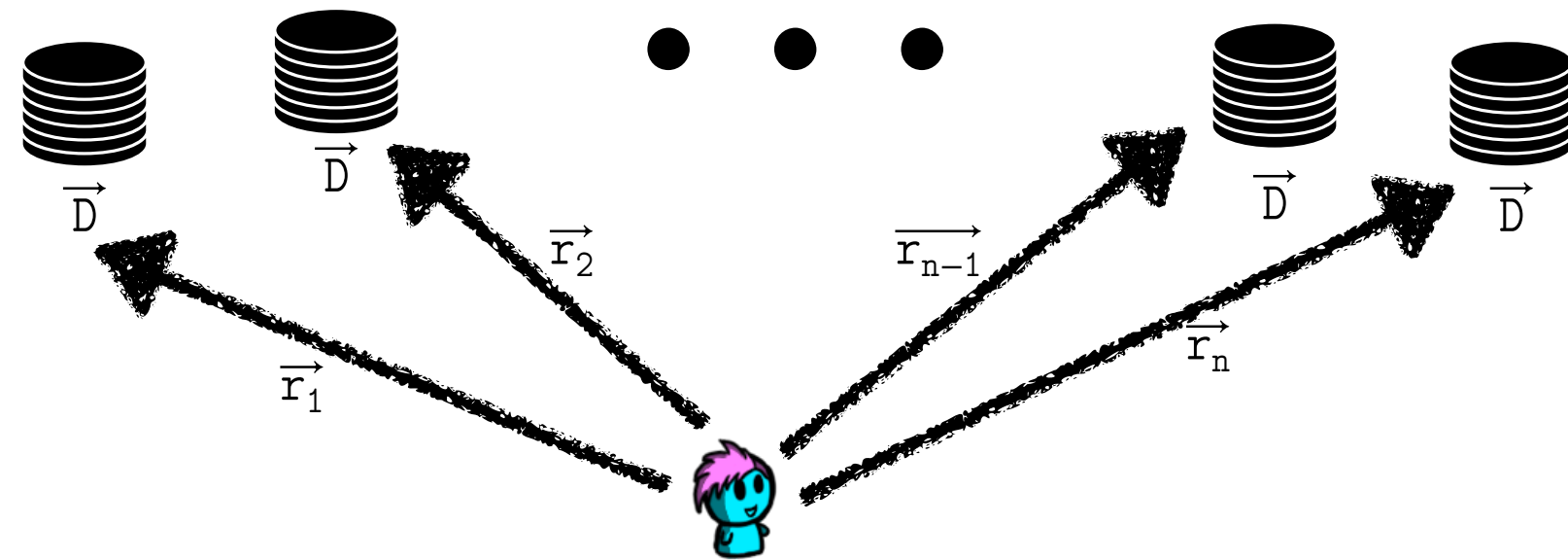
What is the download cost?

IT-Private Information Retrieval



What could go wrong?

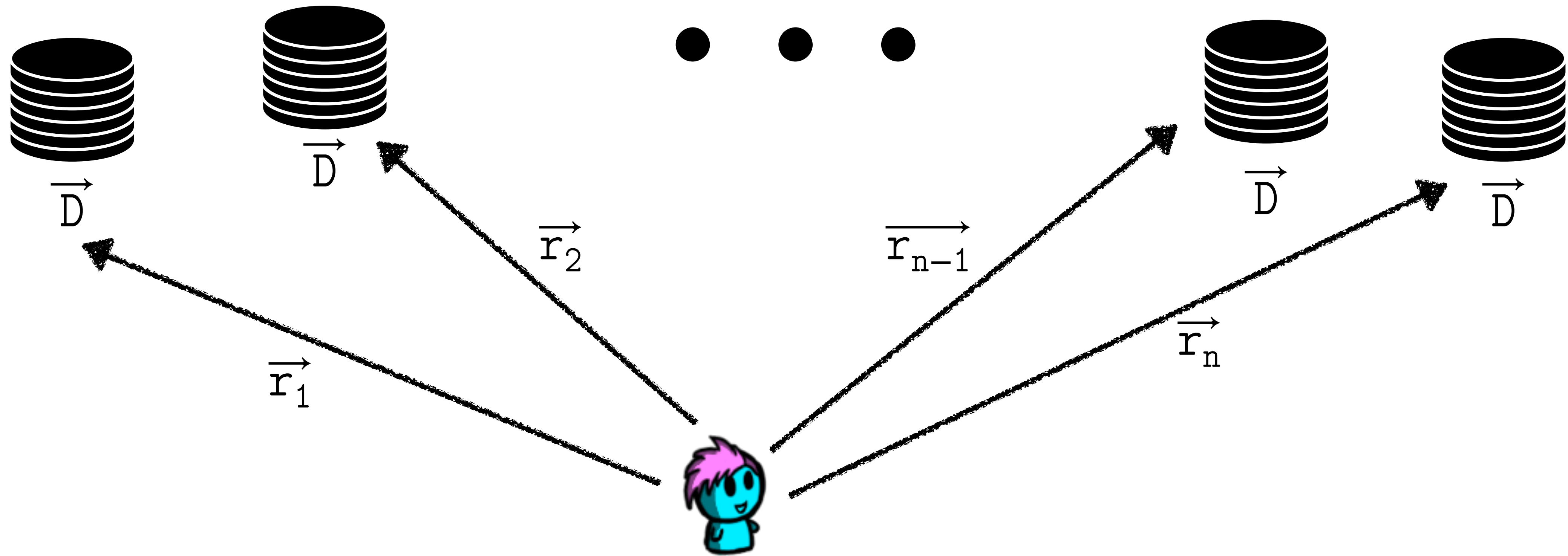
IT-Private Information Retrieval



How to protect against servers that do not respond?

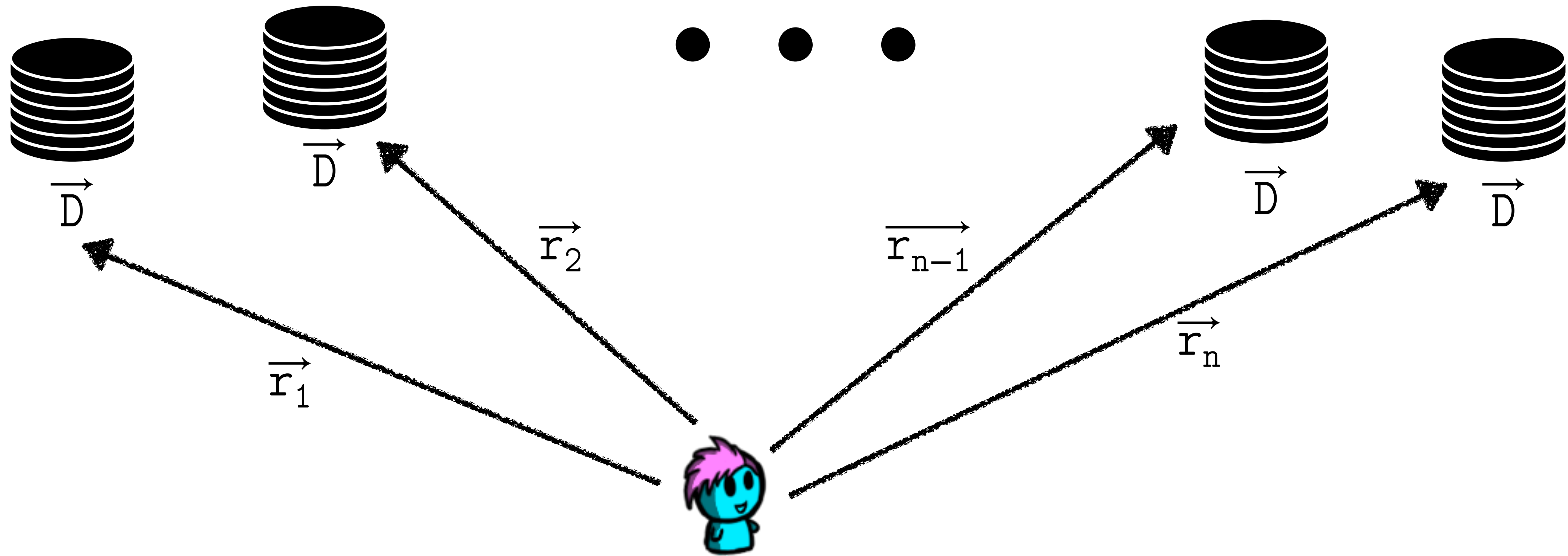
Perhaps we could use a different type of secret sharing?

IT-Private Information Retrieval



What are the downsides of this protocol?

IT-Private Information Retrieval



Can we use Distributed Point Functions here?

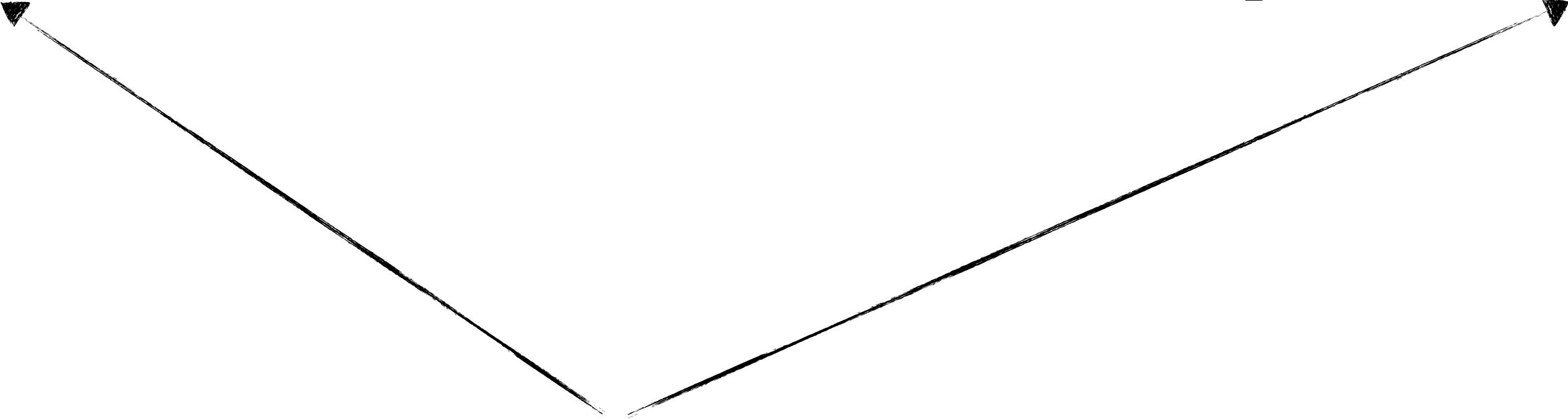
IT-Private Information Retrieval

x_{00}	x_{10}	x_{20}	x_{30}	x_{40}	x_{50}	0
x_{01}	x_{11}	x_{21}	x_{31}	x_{41}	x_{51}	0
x_{02}	x_{12}	x_{22}	x_{32}	x_{42}	x_{52}	0
x_{03}	x_{13}	x_{23}	x_{33}	x_{43}	x_{53}	0
x_{04}	x_{14}	x_{24}	x_{34}	x_{44}	x_{54}	0
x_{05}	x_{15}	x_{25}	x_{35}	x_{45}	x_{55}	0

$$q_1 = [0, 3, 0, 5, 3, 5]$$

x_{00}	x_{10}	x_{20}	x_{30}	x_{40}	x_{50}	0
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x_{05}	x_{15}	x_{25}	x_{35}	x_{45}	x_{55}	0

$$q_1 = [0, 3, 6, 5, 3, 5]$$



IT-Private Information Retrieval

x ₀₀	x ₁₀	x ₂₀	x ₃₀	x ₄₀	x ₅₀	0
x ₀₁	x ₁₁	x ₂₁	x ₃₁	x ₄₁	x ₅₁	0
x ₀₂	x ₁₂	x ₂₂	x ₃₂	x ₄₂	x ₅₂	0
x ₀₃	x ₁₃	x ₂₃	x ₃₃	x ₄₃	x ₅₃	0
x ₀₄	x ₁₄	x ₂₄	x ₃₄	x ₄₄	x ₅₄	0
x ₀₅	x ₁₅	x ₂₅	x ₃₅	x ₄₅	x ₅₅	0

x ₀₀	x ₁₀	x ₂₀	x ₃₀	x ₄₀	x ₅₀	0
x ₀₁	x ₁₁	x ₂₁	x ₃₁	x ₄₁	x ₅₁	0
x ₀₂	x ₁₂	x ₂₂	x ₃₂	x ₄₂	x ₅₂	0
x ₀₃	x ₁₃	x ₂₃	x ₃₃	x ₄₃	x ₅₃	0
x ₀₄	x ₁₄	x ₂₄	x ₃₄	x ₄₄	x ₅₄	0
x ₀₅	x ₁₅	x ₂₅	x ₃₅	x ₄₅	x ₅₅	0

$\vec{q} = [0 \dots s]^r$ such that $\vec{q}[j] = 0$

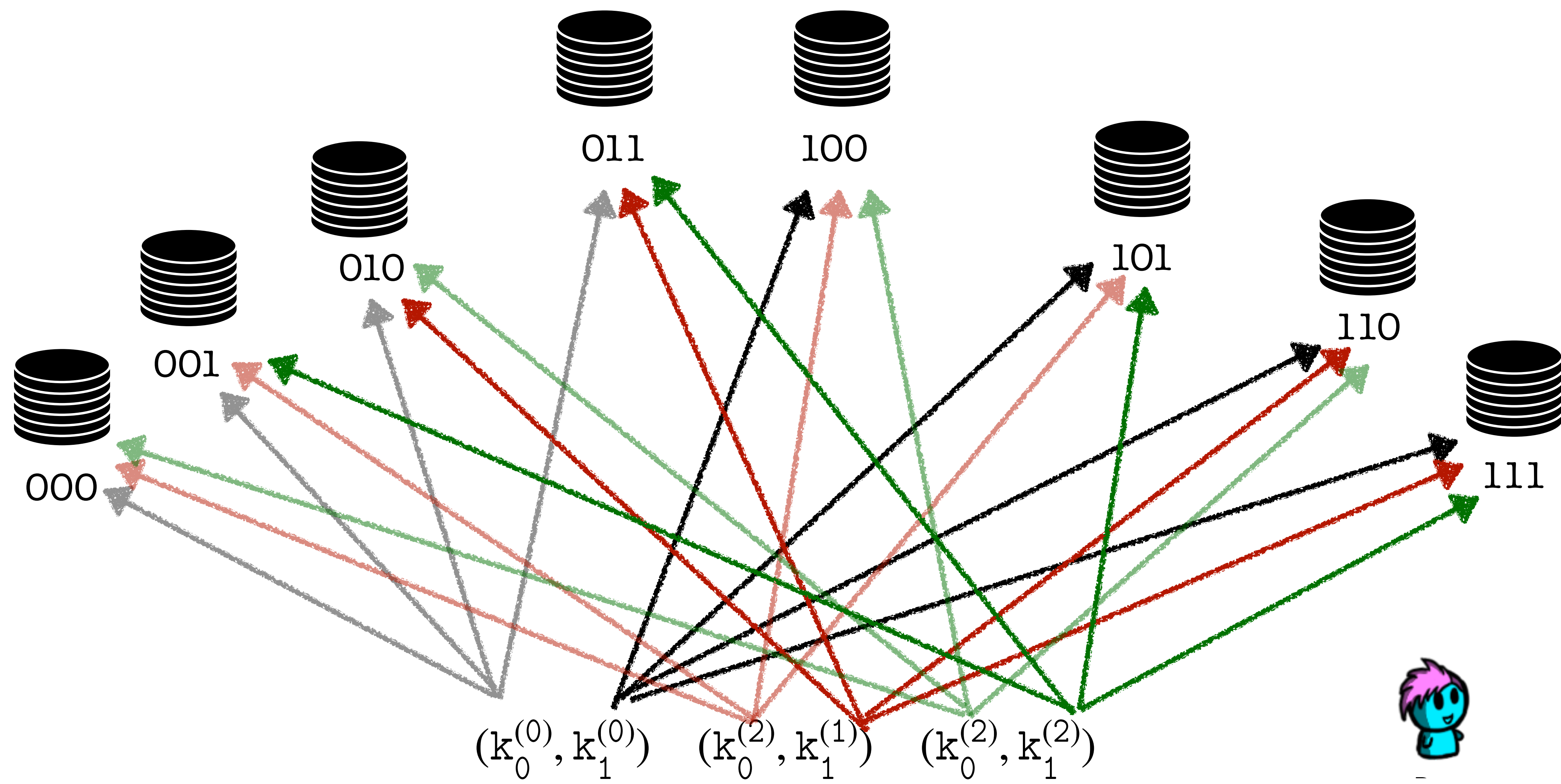
Next user selects a permutation: $\sigma: [0 \dots s] \rightarrow [0 \dots s]$

$$\text{query}_k = q + \sigma(k) \cdot \vec{e}_j$$

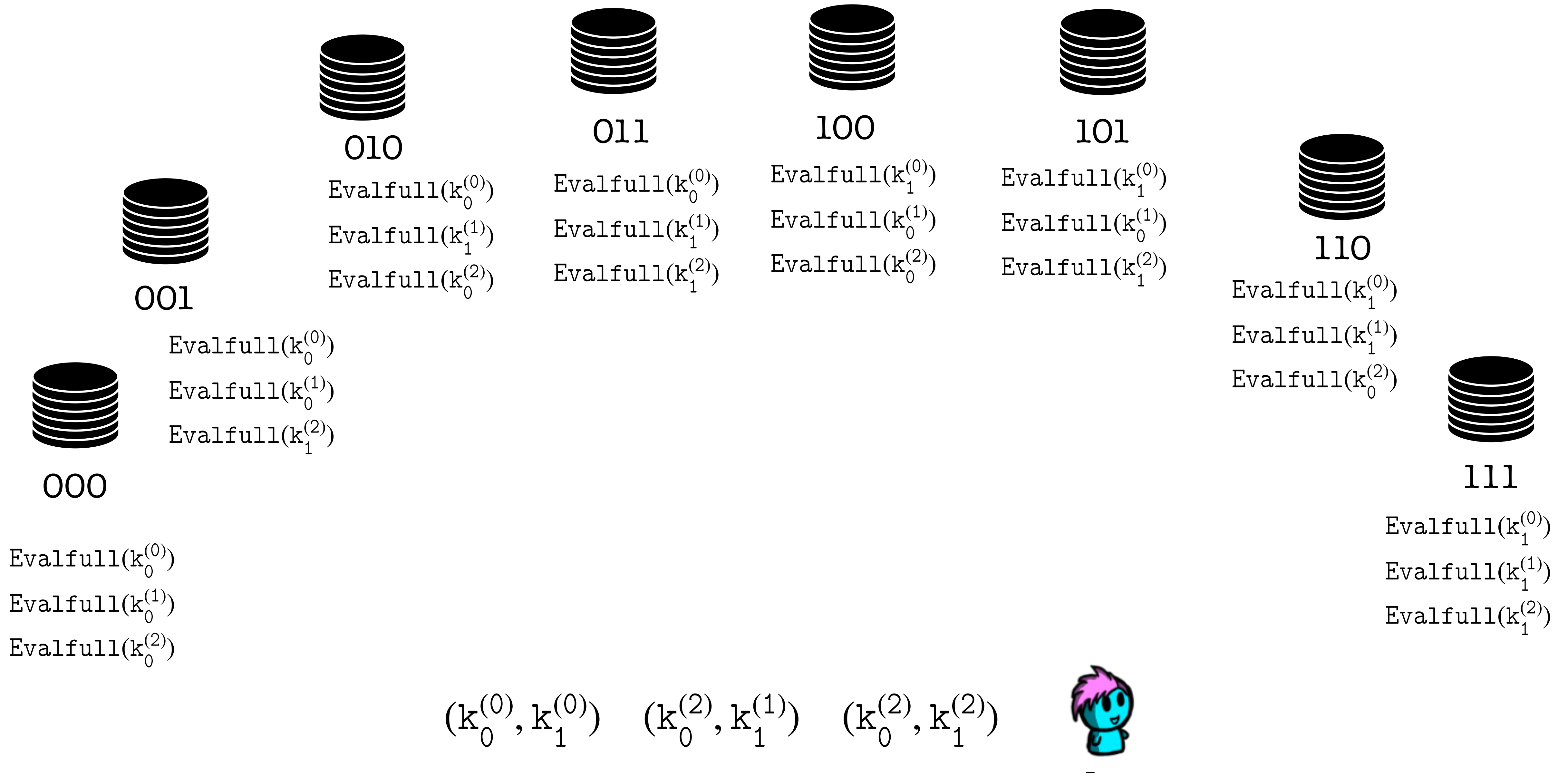
$$\text{response}_k = \bigoplus_{i=1}^r \vec{D}_i[\text{query}_k[i]]$$

User computes $\vec{D}_j[\sigma(k)] = \text{response}_k \oplus \text{response}_0 \quad k = 1, \dots, s$

IT-Private Information Retrieval



IT-Private Information Retrieval



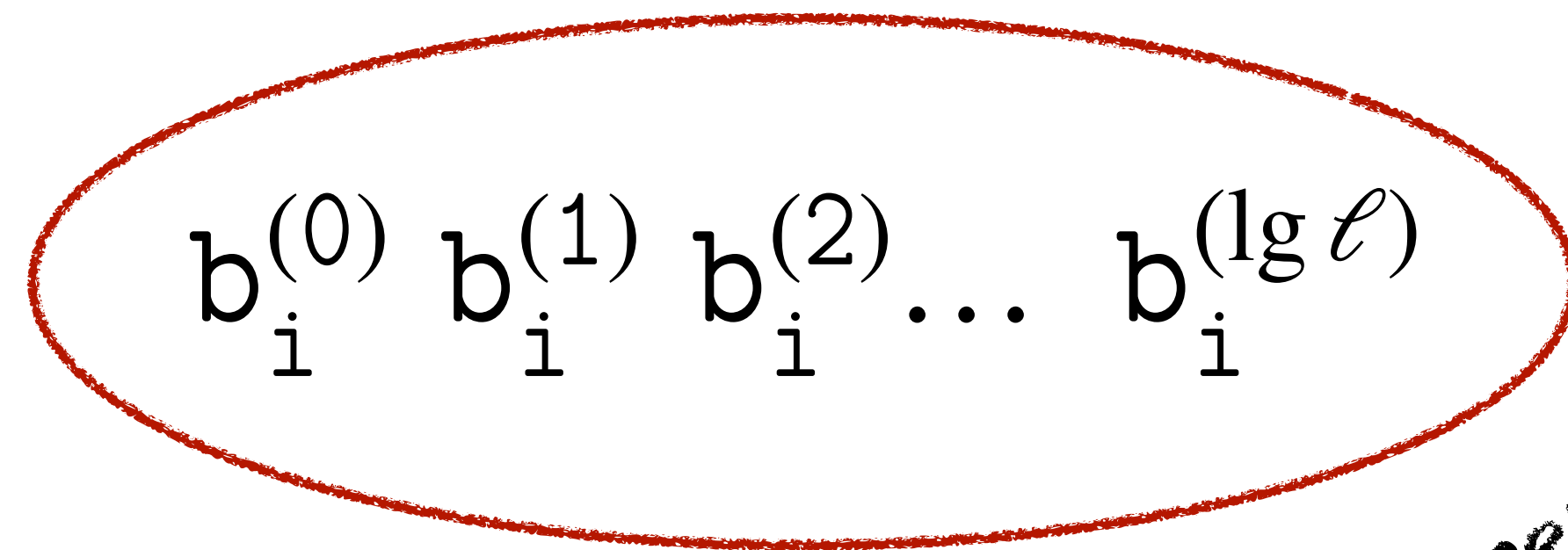
Theorem about achieving Robustness

For any 1-private, (2,2)-PIR protocol, with communication cost X , one can construct a 1-private $(2, \ell)$ PIR protocol with communication cost $X \cdot \lg \ell$

$$(q_0^{(0)}, q_1^{(0)}) \leftarrow \text{Query}(1^\lambda, j)$$

$$(q_0^{(1)}, q_1^{(1)}) \leftarrow \text{Query}(1^\lambda, j)$$

$$(q_0^{(\lg \ell)}, q_1^{(\lg \ell)}) \leftarrow \text{Query}(1^\lambda, j)$$


$$b_i^{(0)} \ b_i^{(1)} \ b_i^{(2)} \ \dots \ b_i^{(\lg \ell)}$$

binary representation of i

Robustness in PIR

What is a robust Private Information Retrieval?

A PIR protocol that allows a few parties to fail

How can we achieve robustness?

Theorem about achieving Robustness

$$(q_0^{(0)}, q_1^{(0)}) \leftarrow \text{Query}(1^\lambda, j)$$

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$$(q_0^{(\lg \ell)}, q_1^{(\lg \ell)}) \leftarrow \text{Query}(1^\lambda, j)$$

$$b_i^{(0)} \ b_i^{(1)} \ b_i^{(2)} \dots \ b_i^{(\lg \ell)}$$

binary representation of i

$$q_0^{(0)}, q_0^{(1)}, \dots, q_0^{(\lg \ell)}$$



$$q_{b_i^{(0)}}^{(0)}, q_{b_i^{(1)}}^{(1)}, \dots, q_{b_i^{(\lg \ell)}}^{(\lg \ell)}$$



Theorem about achieving Robustness

$$q_0^{(1)}, q_1^{(1)} \leftarrow \text{Query}(1^\lambda, j)$$

$$q_0^{(2)}, q_1^{(2)} \leftarrow \text{Query}(1^\lambda, j)$$

$$(q_0^{(1)}, q_0^{(2)})$$



00

$$(q_0^{(1)}, q_1^{(2)})$$



01

$$(q_1^{(1)}, q_0^{(2)})$$



10

$$(q_1^{(1)}, q_1^{(2)})$$



11

Theorem about achieving Robustness

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Theorem about achieving Robustness

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11

Theorem about achieving Robustness

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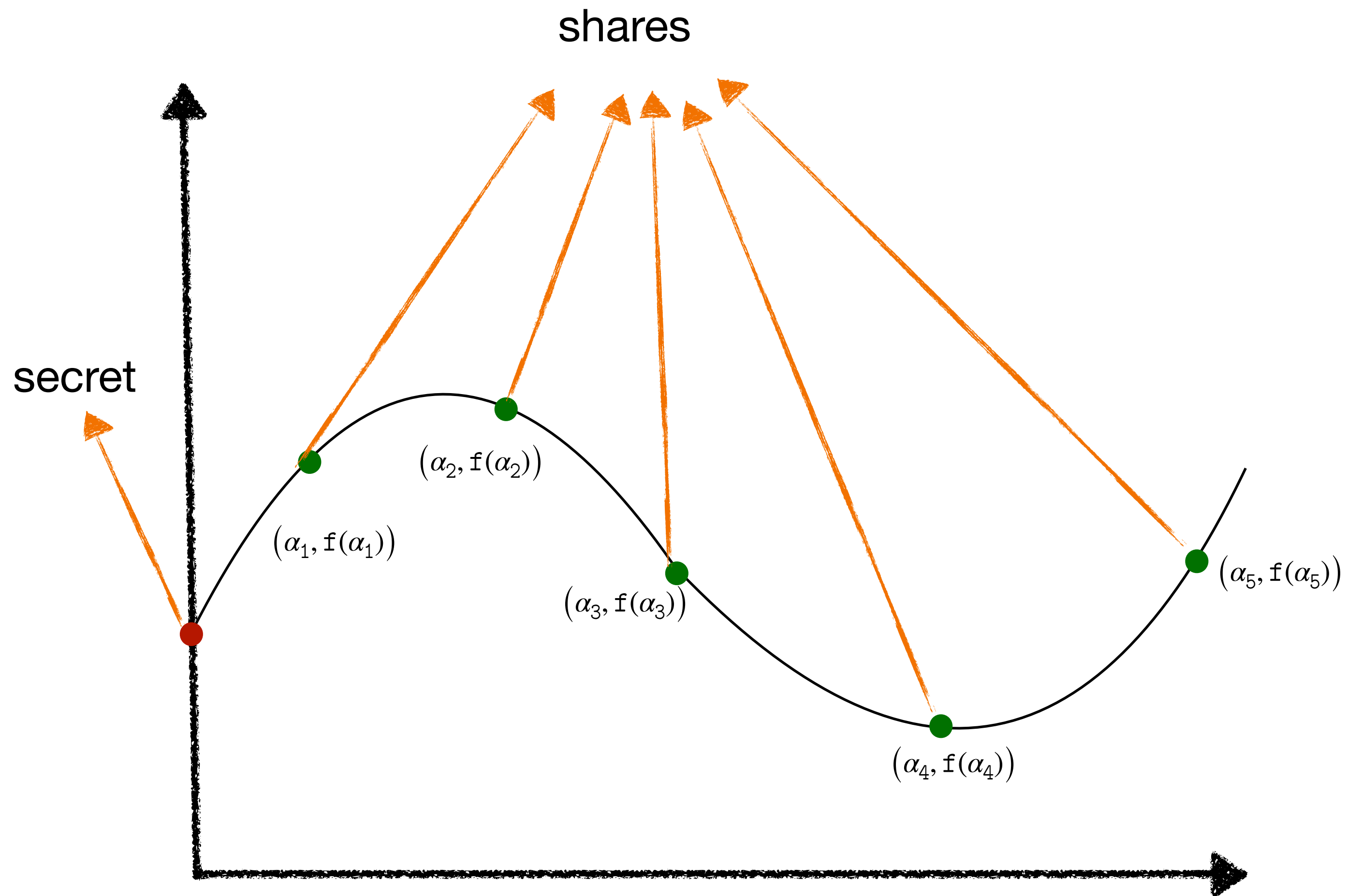
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$$(q_1^{(1)}, q_1^{(2)})$$



11

Shamir Secret Sharing



Properties

Linearity

$$f(x) = s_1 + a_1x + a_2x^2 + \dots + a_tx^t$$

$$g(x) = s_2 + b_1x + b_2x^2 + \dots + b_tx^t$$

$$h(x) = \alpha \cdot f(x) + \beta \cdot g(x)$$

constant term is: $\alpha \cdot s_1 + \beta \cdot s_2$

each party i can locally compute:

$$h(i) = \alpha \cdot f(i) + \beta \cdot g(i)$$

Shamir Secret Sharing

Properties

Linearity

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Why is linearity an important property?

What other important property does Shamir's Secret Sharing have?

Computational PIR using Fully Homomorphic Encryption

What is Fully Homomorphic Encryption?

$$(pk, sk) \leftarrow \text{KeyGen}(1^\lambda)$$

$$\text{Enc}(pk, x_0) + \text{Enc}(pk, x_1) = \text{Enc}(pk, x_0 + x_1)$$

$$\text{Enc}(pk, x_0) \cdot \text{Enc}(pk, x_1) = \text{Enc}(pk, x_0 \cdot x_1)$$

How can FHE can be used in PIR?

Keyword PIR

What do you think is Keyword-Based PIR?

We have assumed that the client knows the exact database index of the record they are looking for.

That is not always useful.

You would rather want: (key , value)

Two Hashes

A regular SHA256 Hash of 32 bytes

A Truncated Hash

$(\text{keyword}, \text{value}) = (w, v)$

$$V(w) = H(w) \parallel v$$

Keyword based PIR

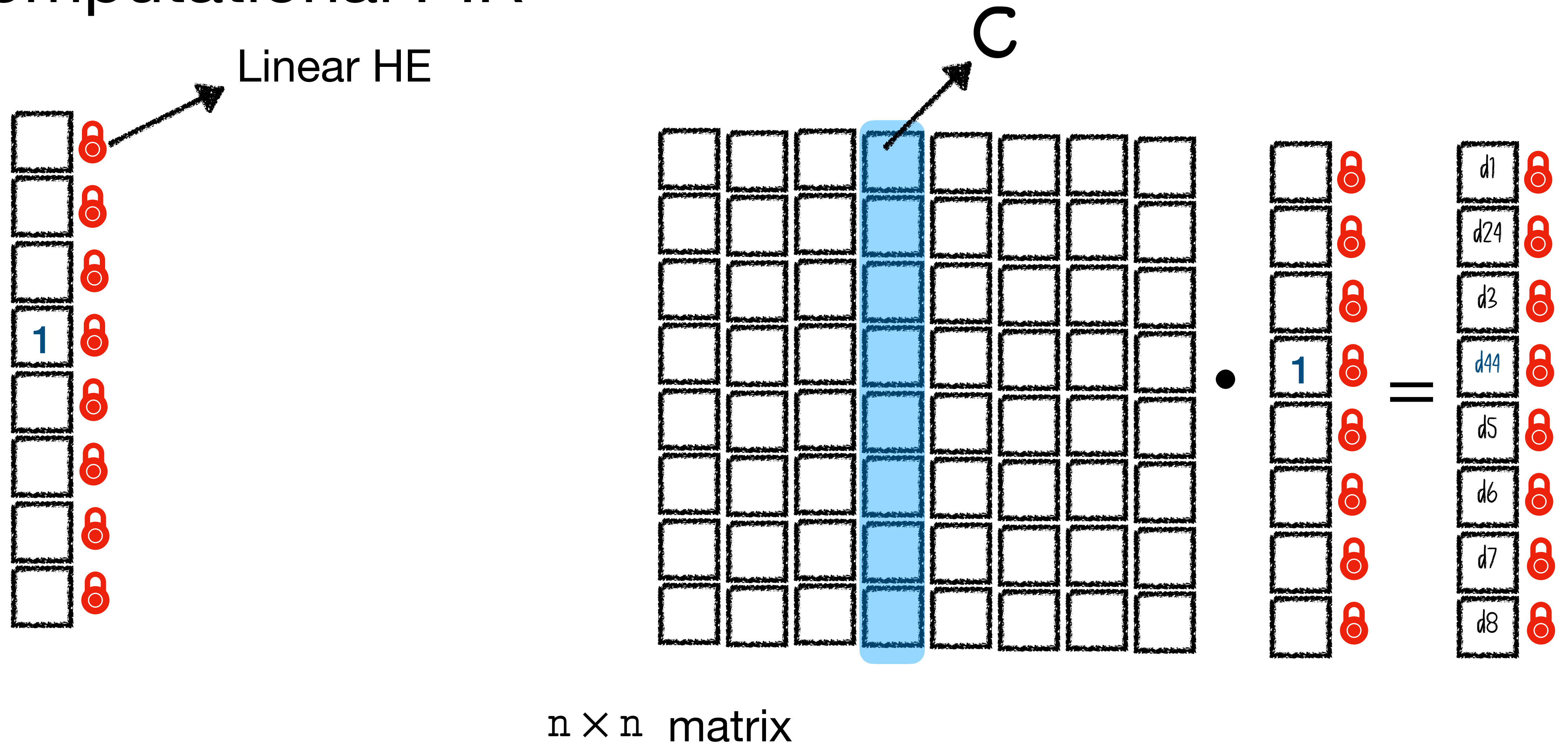
$$(k_0, k_1) \leftarrow \text{Gen}(r, i, 1)$$

$$(k_0, k_1) \leftarrow \text{Gen}(2^d, H_d(w), 1)$$

$$\bigoplus_{j=1}^r D[j] \mid_{\text{Eval}(b, k_b, j)=1}$$

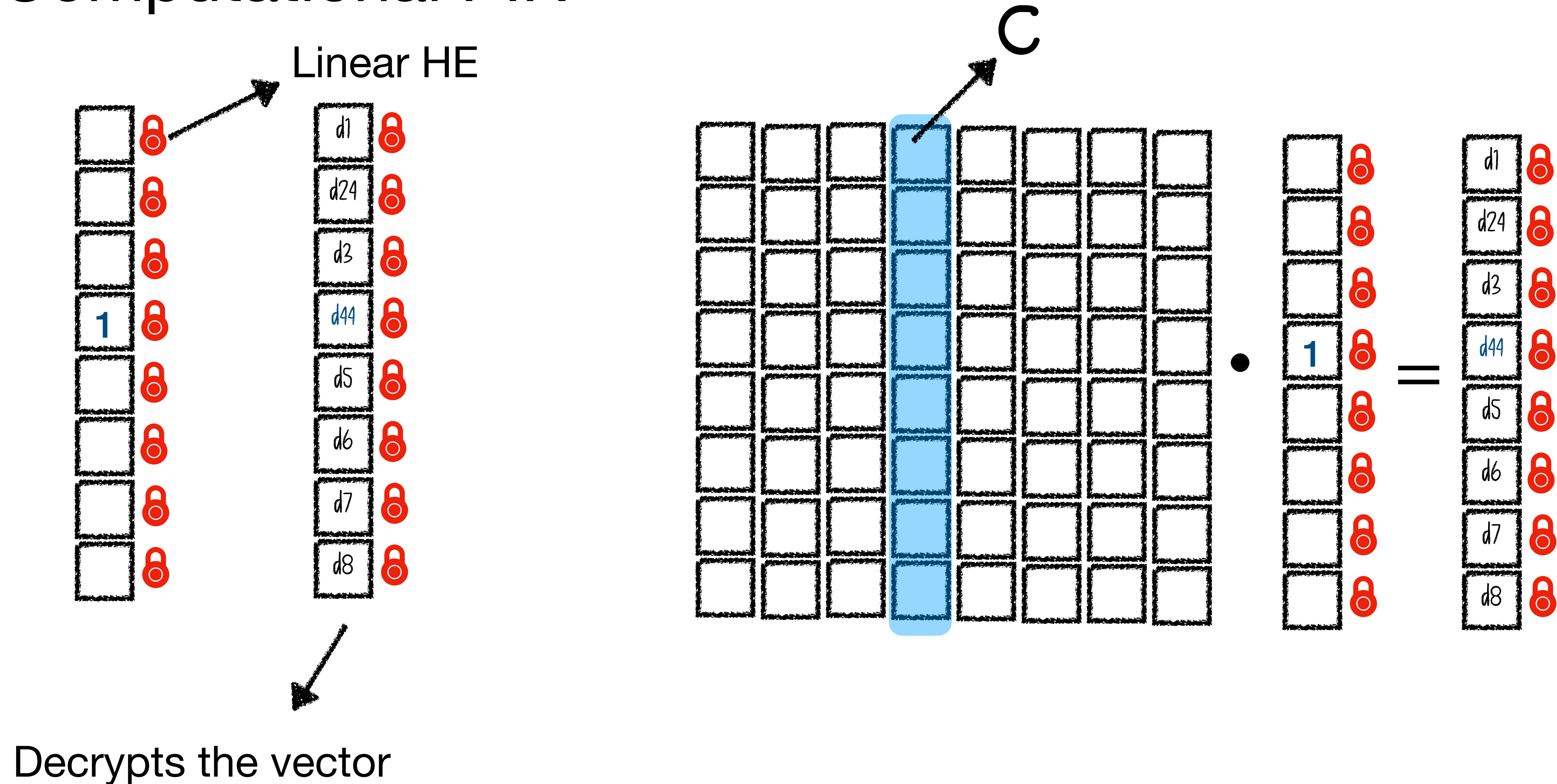
$$\bigoplus_{w \in \text{keywords}} V(w) \mid_{\text{Eval}(b, k_b, H_d(w)=1)=1}$$

Computational PIR

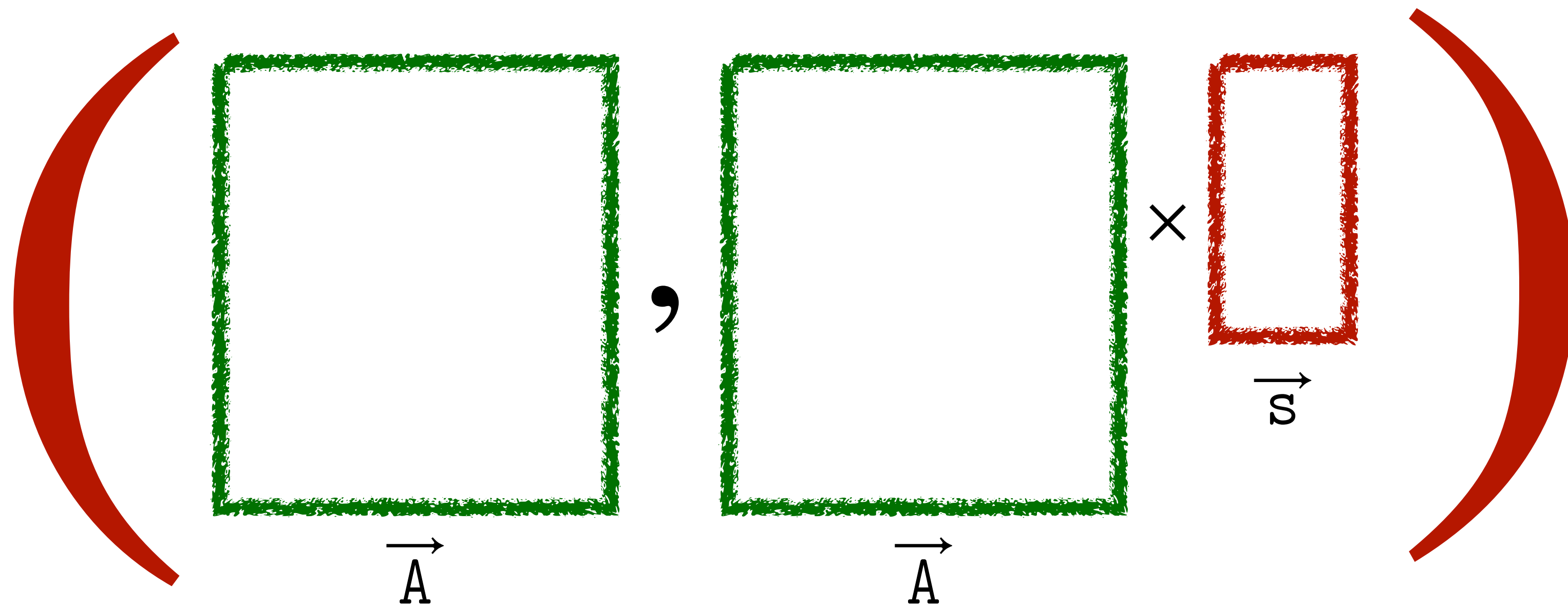


The client wants something in column C

Computational PIR



Learning with Errors Problem

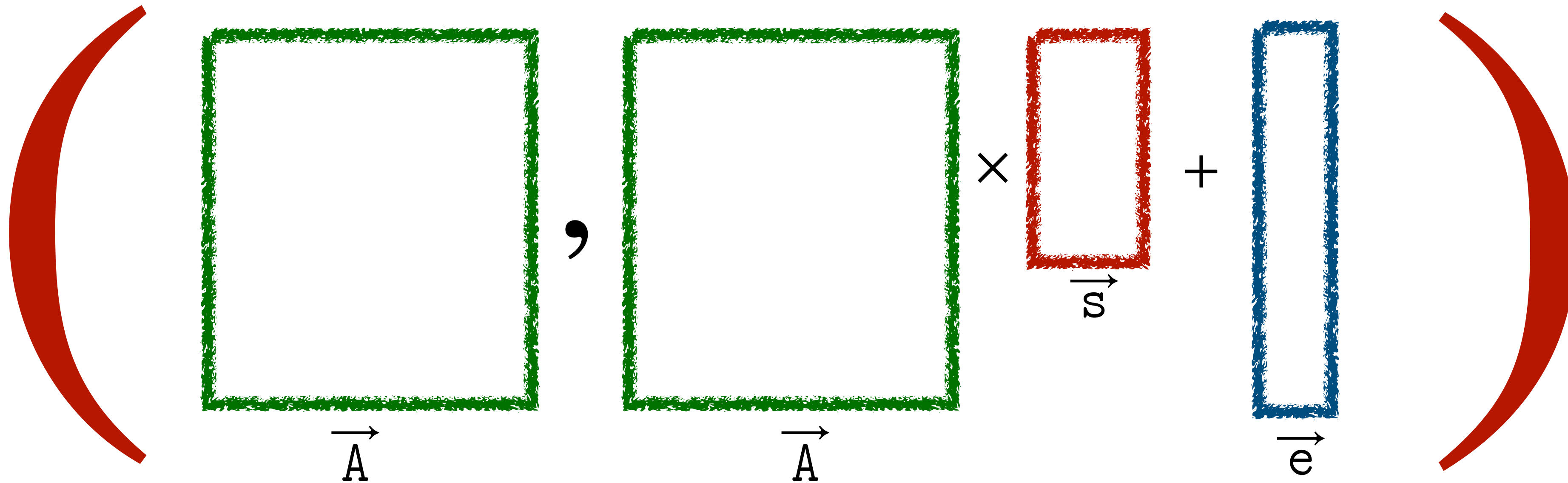


Can we find $\vec{s}$?

Yes, solve the system of linear equations!

Learning with Errors Problem

Hard to distinguish this pair from a random pair



Can we find $\vec{s}$?

We believe the answer is NO!

Learning with Errors Problem

The diagram illustrates the Learning with Errors (LWE) problem. It features a large red left parenthesis on the left and a large red right parenthesis on the right. Inside, there are two green squares representing matrices, each with a vector $\vec{A}$ below it. A comma is placed between these two squares. To the right of the second square is a multiplication symbol $\times$, followed by a red rectangle representing a secret vector $\vec{s}$ with its label below. This is followed by a plus sign $+$ and a blue rectangle representing an error vector $\vec{e}$ with its label below. The entire expression is enclosed in the red parentheses.

$$\left(\vec{A}, \vec{A} \times \vec{s} + \vec{e} \right)$$

How can LWE be used to perform encryption?

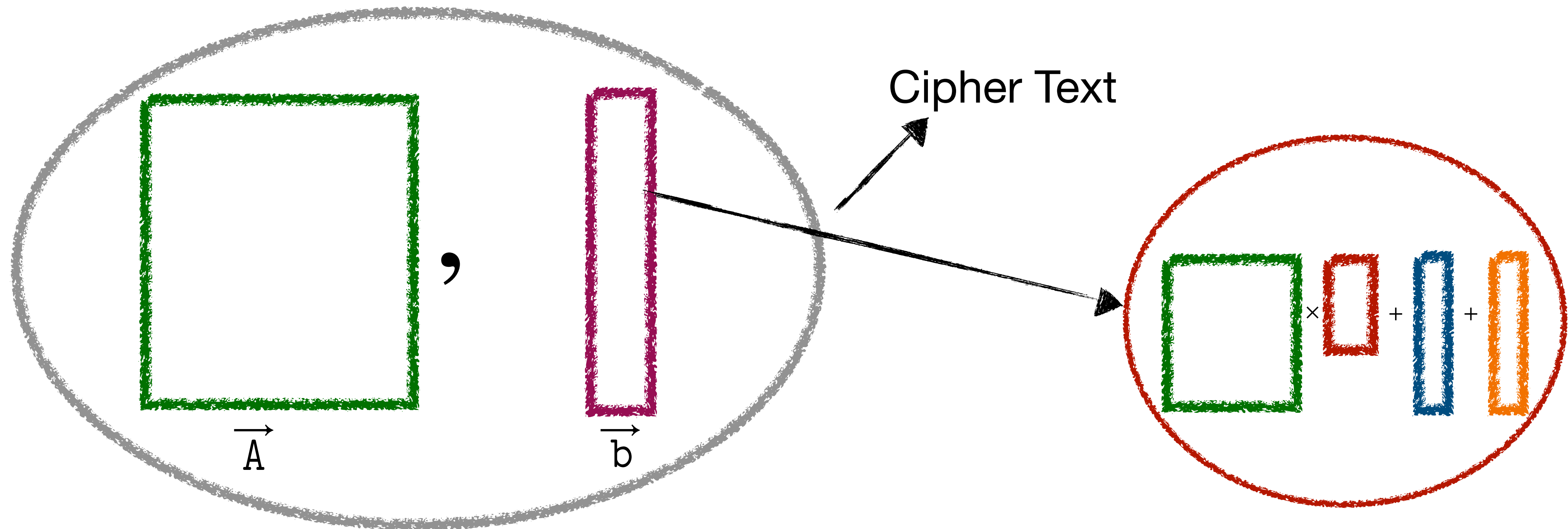
Learning with Errors Problem

The diagram illustrates the Learning with Errors (LWE) problem. It features a large red left parenthesis on the left and a large red right parenthesis on the right. Inside, the equation is represented by colored rectangles and mathematical symbols. Two green rectangles, each labeled $\vec{A}$ below it, are separated by a comma. The first green rectangle is followed by a multiplication symbol $\times$ and a red rectangle labeled $\vec{s}$ below it. This is followed by a plus sign $+$ and a blue rectangle labeled $\vec{e}$ below it. Another plus sign $+$ follows, then an orange rectangle labeled "message" below it. The entire expression is enclosed in the red parentheses.

$$\left(\vec{A}, \vec{A} \times \vec{s} + \vec{e} + \text{message} \right)$$

How can LWE be used to perform encryption?

Learning with Errors Problem



How can LWE be used to perform decryption?

$$\vec{b} - \vec{A} \cdot \vec{s} \approx \text{message}$$

Learning with Errors Problem

We assume the following:

q : be a modulus (typically a power of 2) n : be the dimension

$s \in \mathbb{Z}_q^n$: be a secret vector χ : be an error distribution over $\mathbb{Z}_q$

$$(\vec{a}, b = \langle \vec{a}, \vec{s} \rangle + e) \bmod q$$

Problem: Given many independent samples $(\vec{a}_i, b_i)$ distinguish them from $(\mathbb{Z}_q^n \times \mathbb{Z}_q)$

Alternatively, find the secret vector $\mathbf{s}$

Properties of Regev Encryption

The matrix A is independent of the encrypted message

The scheme is linearly homomorphic

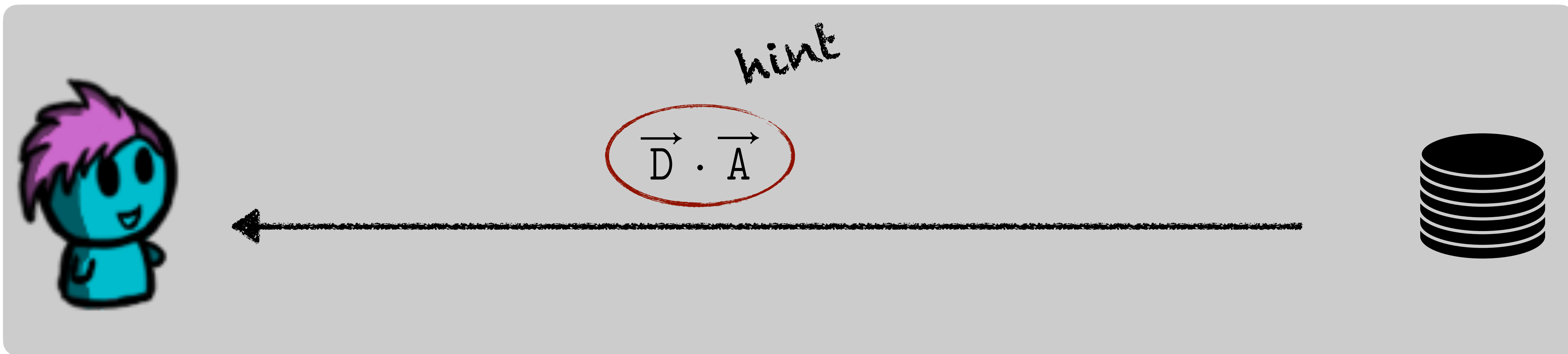
$(\vec{A}, \vec{b})$ encrypts $\vec{e}_{i*}$

$(\vec{D} \cdot \vec{A}, \vec{D} \cdot \vec{b})$ encrypts $\langle \vec{D}, \vec{e}_{i*} \rangle$

Decrypting a single element of the message requires reading a single row of $A || b$

C-PIR Protocol

Preprocessing



Online



Client-Server ORAMs



Outsources the storage to Bob

Of course, Alice will encrypt the database

But, Bob can of course, observe the access patterns!